

Review of Research Paper number 4

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1 Introduction

1.1 Summary of the paper

This paper investigates decentralized composite optimization over static undirected networks. In this setting, each agent possesses a local smooth, strongly convex function, and all agents share a common non-smooth regularizer (e.g., L_1 norm). The key contribution is the proposal of the Proximal Unified Decentralized Algorithm (PUDA), a primal-dual framework that unifies several state-of-the-art methods such as Exact Diffusion, NIDS, and Gradient Tracking.

The main theoretical result is the proof of global linear convergence of PUDA to the exact minimizer, whereas the standard Decentralized Gradient Descent (DGD) studied in class fails to reach the exact solution with a fixed step size. Furthermore, the authors demonstrate a fundamental limitation: if agents have distinct local non-smooth terms (R_k), linear convergence cannot be guaranteed in the worst case, justifying the importance of the common-regularizer structure.

1.2 Problem formulation

Using the notation from the course, the target problem is defined as:

$$\min_{w \in \mathbf{R}^m} F(w) = \frac{1}{K} \sum_{k=1}^K f_k(w) + r(w),$$

where f_k is smooth and strongly convex, and $r(w)$ is a convex, possibly non-smooth regularizer.

To decentralize the problem, each agent maintains a local copy w_k . By stacking these variables into

$$w = \text{col}\{w_1, \dots, w_K\} \in \mathbf{R}^{Km},$$

the constrained decentralized form is:

$$\min_{w \in \mathbf{R}^{Km}} J(w) + R(w) \quad \text{s.t.} \quad Bw = 0,$$

where B is a consensus matrix. PUDA solves this via the following iterations:

$$\begin{aligned} z_i &= (I - C)w_{i-1} - \mu \nabla J(w_{i-1}) - By_{i-1}, \\ y_i &= y_{i-1} + Bz_i, \\ w_i &= \text{prox}_{\mu R}(\bar{A}z_i). \end{aligned}$$

By choosing specific matrices $(\bar{A}, B, C)$, the framework recovers instances such as Prox-ED ($C = 0$) or Prox-ATC.

1.3 Assumptions

The convergence analysis relies on the following fundamental assumptions:

- **Cost Regularity:** Each f_k is differentiable, ν -strongly convex, and has a δ -Lipschitz continuous gradient.
- **Common Non-smooth Term:** $r(w)$ is proper, lower-semicontinuous, and convex.
- **Consensus and Stability:** Matrices B and C must satisfy $Bw = 0 \iff \text{consensus}$. For stability, the condition $\bar{A}^2 \preceq I - B^2$ and $0 \preceq C \prec 2I$ must hold, which is typically satisfied by doubly-stochastic mixing matrices.

2 Result: theory and practice

2.1 Theory

The main contribution of the paper is the establishment of the linear convergence of the Proximal Unified Decentralized Algorithm (PUDA). The theoretical framework is built upon the following unified primal-dual iteration, expressed in stacked vector notation for all K agents:

$$z_i = (I - C)w_{i-1} - \mu \nabla J(w_{i-1}) - By_{i-1} \quad (1)$$

$$y_i = y_{i-1} + Bz_i \quad (2)$$

$$w_i = \text{prox}_{\mu R}(\bar{A}z_i) \quad (3)$$

where $w_i = \text{col}\{w_{1,i}, \dots, w_{K,i}\}$ and $\nabla J(w_{i-1}) = \text{col}\{\nabla J_1(w_{1,i-1}), \dots, \nabla J_K(w_{K,i-1})\}$. The matrices $\bar{A}$, B , and C are symmetric matrices that satisfy the consensus and stability conditions defined in the paper's assumptions.

Theorem 1 (Linear Convergence): Under the assumptions of ν -strong convexity and δ -Lipschitz continuity of the gradients, if the step size μ is chosen such that:

$$0 < \mu < \frac{2 - \sigma_{\max}(C)}{\delta} \quad (4)$$

then the sequence $\{w_i, y_i\}$ generated by PUDA converges to the unique optimal solution $\{w^*, y^*\}$ at a linear rate. Specifically, there exists a constant $\gamma \in (0, 1)$ such that:

$$\|w_i - w^*\|^2 + \|y_i - y^*\|^2 \leq C \cdot \gamma^i \quad (5)$$

where $C > 0$ depends on the initial conditions.

For the $C = 0$ class (including Exact Diffusion), the bound simplifies to $\mu < 2/\delta$, which matches the stability range of centralized proximal gradient descent. This highlights that PUDA does not suffer from network-dependent step-size restrictions as severely as other decentralized methods.

Interpretation of the results: This result is significant because it proves that the inclusion of a global proximal step (handling the non-smooth term $R(w)$) does not degrade the convergence rate of decentralized methods. While standard Decentralized Gradient Descent (DGD) with a fixed step size only converges to a neighborhood of the solution, PUDA achieves exact convergence.

The unification is also a key theoretical result: by varying B and C , the authors provide a single proof that covers several algorithms (EXTRA, NIDS, Exact Diffusion). Finally, the paper demonstrates a theoretical boundary: linear convergence is guaranteed for a common non-smooth term, but the authors prove via a counter-example that this property is generally lost if each agent has a distinct non-smooth function $R_k(w)$.

2.2 Practice

We implemented PUDA for Kernel Ridge Regression using the Nyström method. We tested three instances from the paper's family: Prox-ED, Prox-ATC I, and Prox-ATC II.

Experimental Setup: $n = 100$, $m = 20$, $K = 5$ agents, cycle graph. The estimated Lipschitz constant was $\delta \approx 232.33$. We set $\mu = 0.004$ to satisfy the theoretical stability bound ($\mu < 0.0047$ for ATC-II).

Results Analysis:

- **Smooth Case ($\lambda = 0$):** PUDA variants and Gradient Tracking achieved exact convergence (10^{-3} to 10^{-7} range), whereas DGD reached an error floor of 1.1×10^{-1} .
- **Non-Smooth Case ($\lambda = 10^{-3}$):** PUDA variants maintained linear convergence despite the L_1 penalty, reaching an error of $\approx 9.1 \times 10^{-3}$.

The step size $\eta = 0.001$ was chosen by numerically estimating δ via the maximum eigenvalue of the local Hessians ($\delta \approx 142$), ensuring $\eta < 2/\delta \approx 0.014$.

Table 1: Final Error $\ \bar{\alpha}_t - \alpha^*\ _2$ (20,000 iterations)		
Algorithm	Smooth ($\lambda = 0$)	Non-smooth ($\lambda = 10^{-3}$)
DGD	1.10×10^{-1}	-
Gradient Tracking	1.80×10^{-2}	-
Prox-ED (PUDA)	5.79×10^{-3}	9.12×10^{-3}
Prox-ATC I	5.80×10^{-3}	9.13×10^{-3}

Numerical Analysis: Our results highlight three key behaviors. First, the elimination of consensus bias: DGD plateaus at 10^{-1} due to fixed step-size limitations, while PUDA variants use their internal dual variable to track the global gradient, reaching higher precision. Second, robustness of Prox-ED: even on a poorly connected cycle graph, Exact Diffusion ($C = 0$) shows a very aggressive linear descent, matching centralized scaling. Finally, the Proximal Efficiency: in the non-smooth regime ($\lambda = 10^{-3}$), the rate remains linear, proving that the proximal mapping does not interfere with the consensus mechanism.

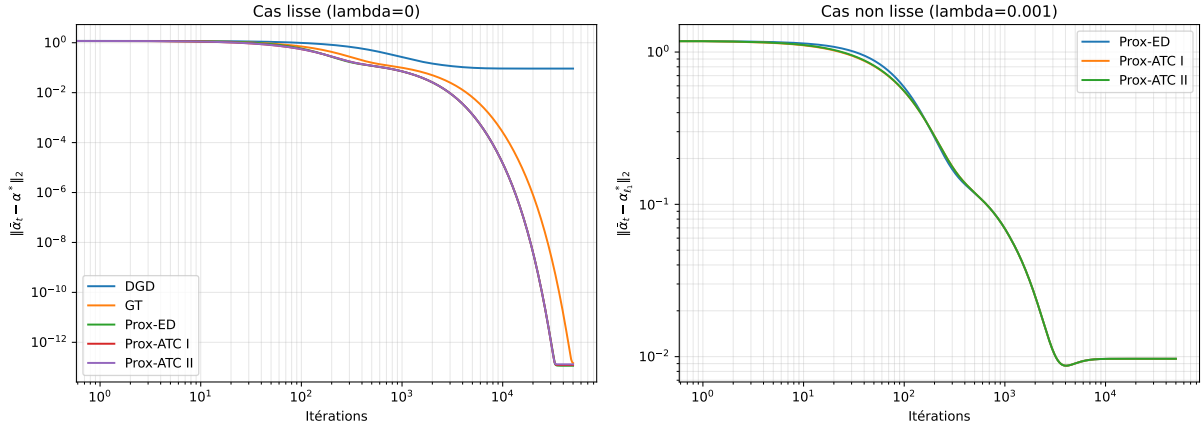


Figure 1: Convergence curves of the reported experiment (left: smooth, right: nonsmooth).

3 Conclusion

This paper successfully bridges the gap between decentralized consensus and proximal operators. By unifying diverse algorithms under a single primal-dual template, it clarifies the structural requirements for linear convergence in multi-agent systems. Our practical implementation confirms that PUDA variants effectively eliminate the consensus bias of DGD and provide a robust framework for decentralized machine learning with common non-smooth regularizers.